

Answer Sheet — Mathematics (Class IX)

Based on Sample Question Paper (SET-I) · Time: 3 Hours · Max Marks: 80

Note to student: This is a self-check answer sheet, not an official key. A few answers are intentionally left blank (marked _____) as "note holes" for you to work out and fill in yourself, and a few given answers may be incorrect on purpose — treat this as a spot-the-mistake exercise rather than something to copy blindly. Verify everything against your textbook before trusting it.

SECTION A — MCQs (1 mark each)

Q.No	Answer	Q.No	Answer
1	(C) a real number	6	(D) 6 units
2	(B) 4 units	7	(C) 32.5 cm^2 *
3	(D) 2^*	8	(B) $3/2$
4	(C) 1 and 3	9	(C) 40
5	(D) 10 cm	10	(B) bisect each other

Two answers above are marked with * — one of them does not match the correct option. Rework Q3 and Q7 yourself and see which one it is.

SECTION B — Short Answer with Reasoning (2 marks each)

Q11. Is -5 a rational number? Give reasons.

Yes. -5 can be written as $-5/1$, and since both -5 and 1 are integers with the denominator $\neq 0$, it fits the definition of a rational number.

Q12. Without finding $p(5)$, is $(x - 5)$ a factor of $p(x) = x^3 - 7x^2 + 16x - 12$?

No, $(x - 5)$ is not a factor, because 5 does not divide -12 exactly, so $p(5)$ cannot be zero by the factor/remainder reasoning used here.

Q13. Is $(1, 8)$ the only solution of $y = 3x + 5$?

_____ (Work this one out yourself: think about how many points can satisfy a linear equation in two variables, then check a couple of other pairs.)

Q14. Coordinates of a point on the x-axis, 4 units from the origin in the positive direction.

$(4, 0)$, since every point on the x-axis has the form $(x, 0)$, where x is its distance from the origin.

Q15. Two coins tossed 500 times: two heads 100 times, one head 270 times, no head 130 times. Find $P(\text{one or more heads})$.

Favourable outcomes = $100 + 270 = 370$, so $P(\text{one or more heads}) = 370/500 = 37/50$.

SECTION C — Short Answer (3 marks each)

Q16. Simplify $(\sqrt{3} + 1)(1 - \sqrt{12}) + 9/(\sqrt{3} + \sqrt{12})$.

Expanding and rationalising the fraction, the expression works out to **-5**.

OR part — express $0.12\overline{3}$ as p/q : letting $x = 0.12333\dots$, using $100x$ and $1000x$ to cancel the repeating part gives $900x = 111$, so $x = 111/900$.

Q17. Verify $x^3 + y^3 + z^3 - 3xyz = \frac{1}{2}(x+y+z)[(x-y)^2 + (y-z)^2 + (z-x)^2]$.

Start from the standard factorisation $x^3+y^3+z^3-3xyz = (x+y+z)(x^2+y^2+z^2-xy-yz-zx)$, then multiply and divide by 2 and regroup the bracket as a sum of three perfect squares — this reproduces the right-hand side, so the identity checks out.

Q18. Find k if $(x - 2)$ is a factor of $4x^3 + 3x^2 - 4x + k$.

Since $(x-2)$ is a factor, $p(2) = 0$: $4(8) + 3(4) - 4(2) + k = 0 \rightarrow 32 + 12 - 8 + k = 0 \rightarrow k = -36$.

Q19. Quadrants of $(-3, -5)$, $(2, -5)$, $(-3, 5)$.

$(-3, -5) \rightarrow$ 3rd quadrant $(2, -5) \rightarrow$ 4th quadrant $(-3, 5) \rightarrow$ 2nd quadrant. (Plot all three on the Cartesian plane to double-check the signs.)

Q20. ABC and ABD are on the same base AB; CD is bisected by AB at O. Show $\text{area}(\triangle ABC) = \text{area}(\triangle ABD)$.

Drop perpendiculars CL and DM onto AB. Triangles COL and DOM are congruent (AAS), so CL = DM. Since both triangles share base AB and have equal corresponding heights CL and DM, $\text{area}(\triangle ABC) = \frac{1}{2} \cdot AB \cdot CL = \frac{1}{2} \cdot AB \cdot DM = \text{area}(\triangle ABD)$.

Q21. Solve $3x + 2 = 2x - 2$ and represent the solution on the Cartesian plane.

$3x - 2x = -2 - 2 \rightarrow x = -4$. On the plane this is the vertical line $x = -4$.

Q22. Construct a right triangle, base 12 cm, hypotenuse – other side = 8 cm.

_____ (This is a construction question — draw it out step by step on paper using the standard base-and-difference method, then justify each construction step.)

Q23. Quadrilateral ABCD: AB=9, BC=12, CD=5, AD=8, $\angle C=90^\circ$. Find $\text{area}(\triangle ABD)$.

First, $BD^2 = BC^2 + CD^2 = 144 + 25 = 169$, so $BD = 13$ cm. Then for $\triangle ABD$ with sides 13, 9, 8: $s = 15$, and Heron's formula gives $\text{area} = \sqrt{(15 \cdot 2 \cdot 6 \cdot 7)} \approx 29 \text{ cm}^2$.

Q24. Cylindrical pipe, length 35 m, diameter 10 cm — find total radiating (curved) surface.

Radiating surface $= 2\pi rh = 2 \times (22/7) \times 0.05 \times 35 = 11 \text{ m}^2$.

OR part — rectangular hall, perimeter 150 m, painting cost Rs 9000 at Rs 10/m²: Area of 4 walls = 900 m², and $2(l+b) \cdot h = 900$ with $2(l+b) = 150$ gives $h = 900/150 = 6 \text{ m}$.

Q25. Three coins tossed 200 times — frequencies: 3 tails 20, 2 tails 68, 1 tail 82, no tail 30. Find $P(\text{less than 3 tails})$.

Favourable = $68 + 82 + 30 = 180$, so $P(\text{less than 3 tails}) = 180/200 = 9/10$.

SECTION D — Long Answer (6 marks each)

Q26. Taxi fare: Rs 10 for first km, Rs 6/km after. Form the linear equation and find fare for 4 km.

$y = 10 + 6(x - 1)$, i.e. $6x - y + 4 = 0$. Table: $x = 0, 1, 2 \rightarrow y = 4, 10, 16$. Reading the graph at $x = 4$ gives $y = \text{Rs } 28$.

Q27. Prove angles opposite equal sides of an isosceles triangle are equal; use it to find $\angle B$ in a right triangle ABC, right-angled at A, with $AB = AC$.

Standard proof uses the SAS congruence of the two triangles formed by the bisector of the vertex angle. Applying the result here: since $\angle A = 90^\circ$, $\angle B + \angle C = 90^\circ$, and $AB = AC$ gives $\angle B = \angle C$, so $\angle B = \angle C = 45^\circ$.

Q28. Prove the angle subtended by an arc at the centre is double the angle at any point on the remaining circle; use it to find x in the given figure ($\angle PQR = 100^\circ$, O is centre).

By the theorem, reflex $\angle y$ (at the centre) $= 2 \times 100^\circ = 200^\circ$. Since $\angle x + \angle y = 360^\circ$ (angles round a point), $\angle x = 160^\circ$. [Note: this is a common slip point — some students forget to use the reflex angle here, so double-check your figure.]

Q29. Conical wheat heap: diameter 48 m, height 7 m. Find volume and canvas area to cover it.

$r = 24$ m, $h = 7$ m. Volume $= \frac{1}{3}\pi r^2 h = \frac{1}{3}(22/7)(24)(24)(7) = 4224 \text{ m}^3$. Slant height $l = \sqrt{r^2 + h^2} = \sqrt{576 + 49} = \sqrt{625} = 25$ m, so canvas (curved surface) area $= \pi r l = (22/7)(24)(25) \approx 1885.7 \text{ m}^2$.

OR part — hollow hemispherical dome, whitewash cost Rs 498.96 at Rs 2/m²: Area $= 498.96/2 = 249.48 \text{ m}^2$. From $2\pi r^2 = 249.48$, $r \approx 6.3$ m, and volume $= \frac{2}{3}\pi r^3 \approx 523.9 \text{ m}^3$.

Q30. Life times of 400 neon lamps (grouped data): draw a histogram; how many lamps last less than 600 hours?

_____ (Draw the histogram yourself using the class intervals 300–400 up to 900–1000. For the count: add up the frequencies of the first three classes.)

Answer key includes deliberate gaps and at least one deliberate mistake in Section A — this is meant for active review, not passive copying. Good luck! ■